

IIT Bombay - National Entrepreneurship Challenge

E-Cell Track Application

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Team: Titan - Bio-Materials & Cleantech

My Motivation

My interest in bio-materials began in our chemical synthesis lab while working with sugarcane bagasse extraction. Seeing kilograms of agricultural waste being burned in neighboring fields while local vendors wrap produce in single-use plastics felt like a massive disconnect. Academic papers write endlessly about green polymers, but very few of those lab formulations ever reach a factory floor.

My motivation is simple: bridge the gap between green chemistry research and commercial packaging. I want to build biodegradable packaging films that can run on existing machinery without forcing manufacturers to buy millions in new equipment. It's a tough engineering and chemical puzzle, but one that has immediate environmental impact.

My Expectations from E-Cell

I expect E-Cell to help us translate lab results into a viable business case. Chemical startups face steep hurdles--regulatory clearances from pollution control boards, testing shelf-life stability, and competing against dirt-cheap petroleum plastics.

Through the NEC platform, I want to learn how to negotiate pilot runs with FMCG companies, evaluate supply chain logistics for agricultural waste collection, and build financial models that account for seasonal raw material price fluctuations.